Machine Learning in Practice I
Lecture 5

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24/10/25

Probability and information theory basics

Information theory basics

Practical **probability theory** (adapted to ML):

- possible outputs of an event: sample space $\omega \in \Omega$
- **features** ($\rightarrow$ random variables): $X : \Omega \rightarrow \mathbb{R}^n$ functions, $x = X(\omega)$
- observe a list of events: $(\omega_1, \omega_2, \ldots, \omega_N)$
- **probability**: $P(X \in U \subset \mathbb{R}^n) = \frac{1}{N} \sum_{i=1}^{N} \delta_{X(\omega_i) \in U}$ $\rightarrow$ **histogram**
- **distribution**: $p_X(x) = P(X \in \{x\})$
- *probability density:* $p_X(x) = \frac{P(X \in [x, x+dx])}{|dx|}$

Indicator function

$$\delta_b = \begin{cases} 1 \text{ if } b = True \\ 0 \text{ if } b = False \end{cases}$$

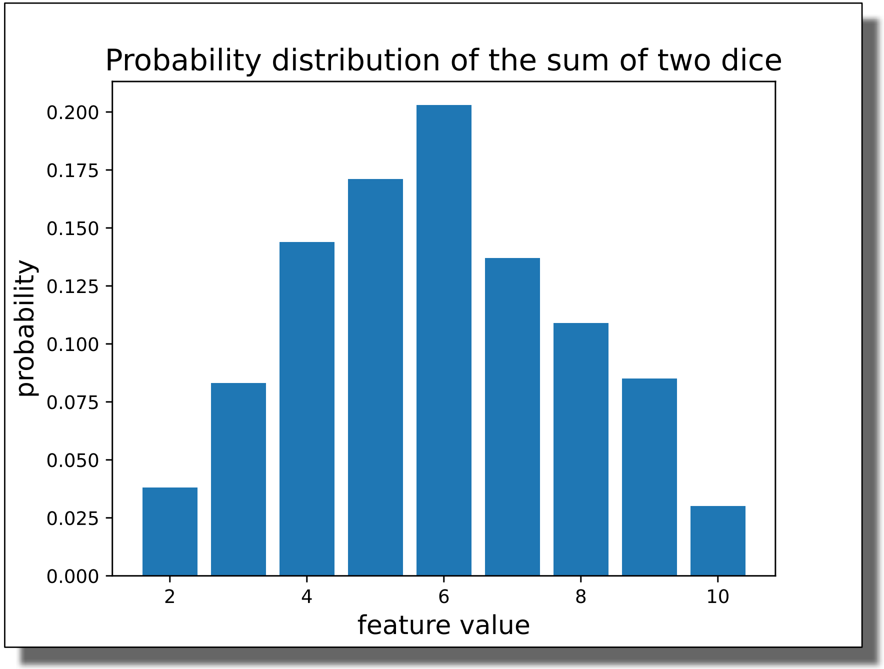
Information theory basics

example of 2 dice thrown N times

```
# generate data
Ndata = 1000
data = np.random.randint(1,6,size=(Ndata,2))
feature = data[:,0] + data[:,1]

# measure histogram
fmax = int(max(feature))
fmin = int(min(feature))
bins = np.zeros(fmax-fmin+1)
for f in feature:
    bins[int(f-fmin)] += 1
bins/=Ndata

# show result
fig,ax = plt.subplots()
ax.bar(np.arange(fmin,fmax+1),bins)
ax.set_xlabel('feature value', fontsize=14)
ax.set_ylabel('probability', fontsize=14)
```



Information theory basics

Joint probability distributions

- the features can have several components → $XY=(X,Y):\Omega\rightarrow\mathbb{R}\times\mathbb{R}$
- their distribution is called joint probability distribution: $p_{XY}(x,y)$
- if we are interested only on one component: $p_X(x)=\sum_{y\in Y} p_{XY}(x,y)$
- for independent features the value of x does not depend on the value of y $p_{XY}(x,y)=p_X(x)\,p_Y(y)$

$$\boxed{p_X(x)=\frac{1}{N}\sum_{i=1}^{N}\delta_{X(\omega)=x}=\frac{1}{N}\sum_{i=1}^{N}\sum_{y\in Y(\Omega)}\delta_{X(\omega)=x;Y(\omega_i)=y}=\sum_{y\in Y(\Omega)}p_{XY}(x,y)}$$

Information theory basics

Information theory: statistical relation of measurements

- assume we have measured a lot $N \to \infty$
- allows simplifications → Stirling formula: $\ln(n!) \approx n \ln n + \ldots$
- measure of the uncertainty / unpredictability / impurity of a probability distribution → **Shannon entropy**

Information theory basics

Shannon entropy: measure of variability of a distribution

- perform N measurements with n possible outputs (bins)
- count the number of results in each bin: $\{N_1, N_2, \ldots, N_n\}$
- how many realizations is possible for a given $\{N_1, N_2, \ldots, N_n\}$?
- Shannon entropy: log of this number over N
- *equivalent to the entropy concept in physics: number of microstate representation of a macrostate*

Information theory basics

Shannon entropy:

- take n bins, N independent measurements
- we obtain $\{N_1, N_2, \ldots, N_n\}$ $\rightarrow$ probability $p_i = \frac{N_i}{N}$
- how many ways can this distribution be realized? *(measurements within a bin are indistinguishable)*
- binomial – multinomial expression

$$\{N_1, N_2\} \rightarrow \binom{N}{N_1} = \frac{N!}{N_1! N_2!} \qquad \{N_1, N_2, \ldots, N_n\} \rightarrow \frac{N!}{N_1! N_2! \ldots N_n!}$$

Information theory basics

Shannon entropy:

- information theory: $N_i \to \infty$ for all parts
- Stirling formula, using $N = \sum_{i=1}^{n} N_i$

$$\log \frac{N!}{N_1! N_2! \ldots N_n!} \approx N \log N - \sum_{i=1}^{n} N_i \log N_i = -N \sum_{i=1}^{n} p_i \log p_i$$

- Shannon entropy:

$$H = \lim_{N_i \to \infty} \frac{1}{N} \log_2 \frac{N!}{N_1! N_2! \ldots N_n!} = -\sum_{i=1}^{n} p_i \log_2 p_i$$

Information theory basics

Shannon entropy: $H = -\sum_{i=1}^{n} p_i \log_2 p_i$

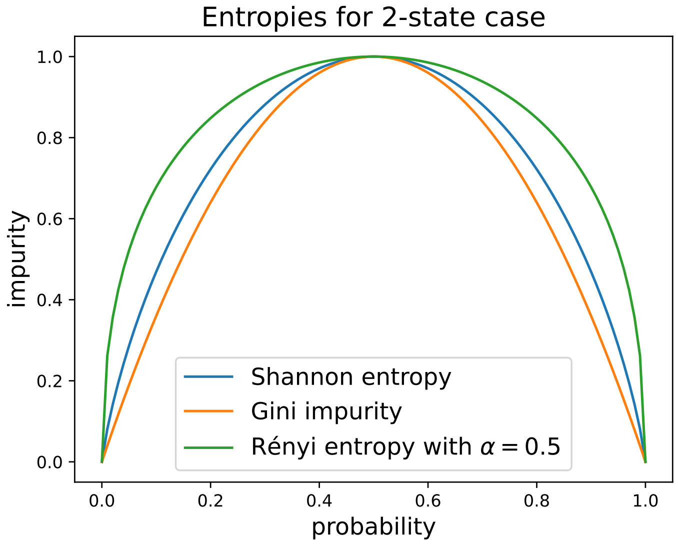
- always positive or zero $H \geq 0$
- minimal for a pure sample $p_1 = 1, p_{i>1} = 0 \Rightarrow H = 0$
- maximal for uniform sample $p_i = \frac{1}{N} \Rightarrow H = \log_2 N$
- convex

$$\boxed{\text{if } E = \sum_i E_i P_i \text{ fixed} \Rightarrow p_i \sim e^{-\beta E_i}}$$

Information theory basics

Other entropy formulae:

- we assumed independence of measurement bins
  what if they are not (non-additive systems)
- general definition using properties
  (positive, convex)
- examples:
  - Rényi-Tsallis entropy $H_\alpha = \frac{1}{1-\alpha} \log\left(\sum_{i=1}^{n} p_i^\alpha\right)$
  - Gini impurity $G = 1 - \sum_{i=1}^{n} p_i^2$



Information theory basics

Shannon entropy and amount of information

- we have n events with probabilities $\{p_1, p_2, \ldots, p_n\}$
- define **information** as the **minimal number of questions** needed to find out which one happened? → *twenty questions game*
- **answer**: Shannon entropy!
  - one question → one "bit" of information
  - Shannon entropy = information content
  - for uniform probabilities logarithmic search, in general Huffmann coding

Information theory basics

Shannon entropy and amount of information – technical slide

- coding: a binary code corresponds an interval in [0,1)

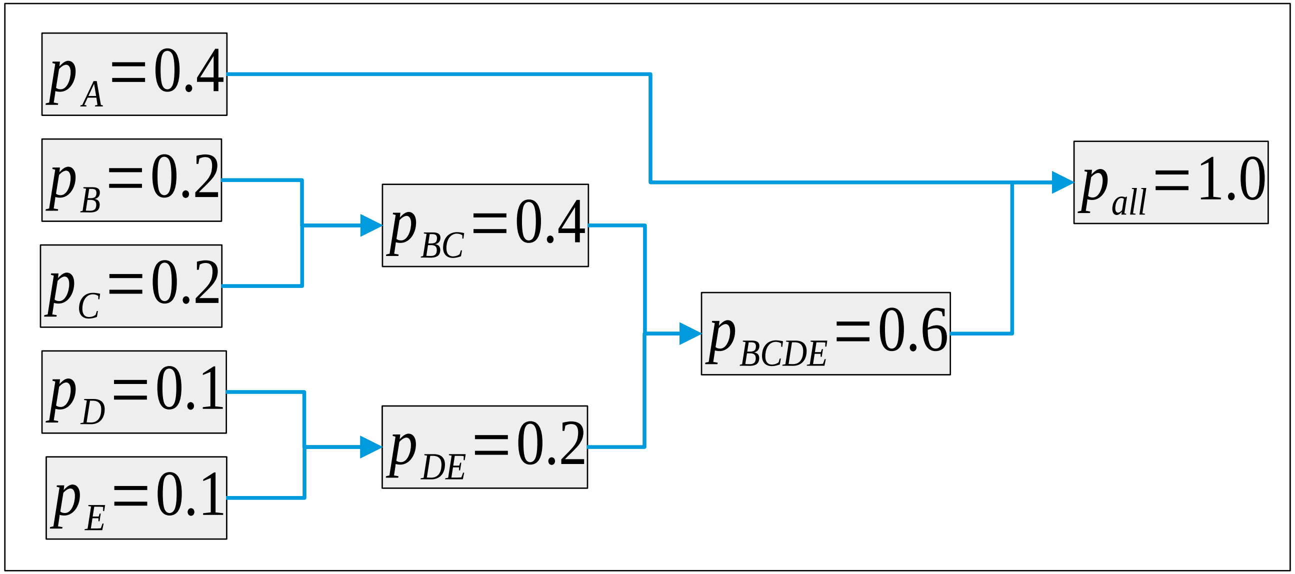
$$(b_1, b_2, \ldots, b_k) \Rightarrow [\sum_{i=1}^{k} b_i 2^{-b_i}, \sum_{i=1}^{k} b_i 2^{-b_i} + 2^{-k}]$$

$$\boxed{101 \Rightarrow (1,0,1) \Rightarrow [\frac{1}{2}+\frac{1}{8}, \frac{1}{2}+\frac{1}{8}+\frac{1}{8}] = [\frac{5}{8}, \frac{6}{8}]}$$

- can be placed non-overlapping if $\sum_{i=1}^{k} 2^{-l_i} \leq 1$ (Kraft–McMillan theorem)
- choose code lengths $l_i = \lfloor -\log_2 p_i \rfloor$ $\rightarrow$ non-overlapping
- average code length $H \leq \sum_{i=1}^{n} p_i l_i < H+1$

Shannon entropy and amount of information

- example for Huffmann coding
- average questions: 0.4+3x0.6=2.2
- Shannon entropy: 2.12



Information theory basics

Joint probability, entropy and mutual information

- if states depend on two variables, the joint probability $p(x,y)$
- for independent variables $p(x,y)=p(x)p(y)$
- the corresponding Shannon entropy

$$H(X,Y)=-\sum_{x\in X,y\in Y}^{n} p(x,y)\log_2 p(x,y) \rightarrow H(X)+H(Y)$$

- measure of independence: mutual information

$$I(X,Y)=H(X)+H(Y)-H(X,Y)$$

Joint probability, entropy and mutual information

- example: 2 bits

```
# Shannon entropy function
def SH(p):
    return -p*np.log2(p+1e-10)

pX = np.sum(p0, axis=1)
pY = np.sum(p0, axis=0)
pXY = np.reshape(p0, -1)
allH = []
for p in [pXY,pX,pY]:
    H = 0
    for pi in np.reshape(p, -1):
        H += SH(pi)
    allH.append(H)
IXY = allH[1] + allH[2] - allH[0]
```

Joint probability, entropy and mutual information

- example: 2 bits

```
# Shannon entropy functio
def SH(p):
    return -p*np.log2(p+1e-1

pX = np.sum(p0, axis=1)
pY = np.sum(p0, axis=0)
pXY = np.reshape(p0, -1)
allH = []
for p in [pXY,pX,pY]:
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    for pi in np.reshape(p, -1
        H += SH(pi)
    allH.append(H)
IXY = allH[1] + allH[2] - allH[0]
```

```
# example: independent case

p0=[[ ¼, ¼ ], [¼, ¼]]

pX = [½, ½]
pY = [½, ½]

# note that pXY = pX*pY

HX = 1
HY = 1
HXY = 2
IXY = 0
```

Joint probability, entropy and mutual information

- example: 2 bits

```
# Shannon entropy functio
def SH(p):
    return -p*np.log2(p+1e-1

pX = np.sum(p0, axis=1)
pY = np.sum(p0, axis=0)
pXY = np.reshape(p0, -1)
allH = []
for p in [pXY,pX,pY]:
    H = 0
    for pi in np.reshape(p, -1
        H += SH(pi)
    allH.append(H)
IXY = allH[1] + allH[2] - allH[0]
```

```
# example: independent case

p0=[[ ¼, ¼ ], [¼, ¼]]

pX = [½, ½]
pY = [½, ½]

# note that pXY = pX*pY

HX = 1
HY = 1
HXY = 2
IXY = 0
```

```
# example: not independent case

p0=[[ 1/3, 1/6 ], [¼, ¼]]

pX = [½, ½]
pY = [7/12, 5/12]

# note that pXY not = pX*pY

HX = 1
HY = 0.98
HXY = 1.96
IXY = 0.02
```

Information theory basics

Probability of the observed sample

- we randomly choose case ***i*** with probability q_i
- what is the probability that we observe N_i events from class ***i***?
- logic of the result:
  - first we choose event 1, next event 2, etc. $\rightarrow \prod_{i=1}^{n} q_i^{N_i}$
  - number of such events is $\dfrac{N!}{N_1!N_2!\dots N_n!}$
  - probability $P=N!\prod_{i=1}^{n}\dfrac{q_i^{N_i}}{N_i!}$

Information theory basics

Probability of the observed sample

- logarithm of the result in the N → ∞ case

$$\log P = N\log N + \sum_{i=1}^{n}\left(N_i \log q_i - N_i \log N_i\right) = N\sum_{i=1}^{n} p_i \log\frac{q_i}{p_i}$$

- Kullback-Leibler divergence: $D(p\|q) = \frac{1}{N}\ln P$

$$\boxed{D(p\|q) = \sum_{i=1}^{n} p_i \ln\frac{q_i}{p_i}}$$

- interpretation: “distance” of probability distributions

END OF LECTURE 5